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Rib-Steered Motor as Solution for Demanding Multi-Lateral Coiled Tubing Drilling Challenges

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Abstract

This paper presents the unique synergy is essential for achieving optimal dogleg severity (DLS) capability, allowing for a single bottom-hole assembly (BHA) that can directionally drill in multiple sections all in a single run. This approach enhances the existing Rib-Steered Motor (RSM) technology by optimizing the wellbore profile with minimal tortuosity.

Conventionally, a positive displacement motor (PDM) is used as the drive type in two different bent configurations to achieve the required DLS in hard carbonate applications. During the drilling process, a high adjustable kick-off (AKO) motor was employed at the beginning of the first lateral, just below the whipstock window or the 4.5-inch liner, to achieve a high DLS of approximately 25 degrees per 100 feet during the build-up section. After this phase, the operation involved pulling out of the hole (POOH) to change the motor for a medium AKO, which was necessary to drill the lateral section with a significantly lower DLS requirement.

After reaching the total depth (TD) of the first lateral, the medium AKO motor BHA was POOH to the second lateral side-tracking depth. Due to the lower DLS expected from this lateral section BHA, side-tracking process took a longer period, and in the worst-case scenario, it failed to initiate the next lateral. This resulted in an additional trip required to use a high AKO motor to provide a higher DLS for sidetracking.

Historically, first-generation RSM was used in the same application but only in the lateral section, with a DLS ranging from 5 to 10 deg/100ft. However, in response to current challenges and application requirements, RSM has been enhanced to achieve a higher DLS capability, targeting 25 deg/100ft.

Prior to the field test, finite element analysis (FEA) was conducted on the second-generation RSM (RSM2) in conjunction with the existing 3.625-inch fixed cutter drill bit design. Multiple iterations of the bit features were simulated to identify the most effective configuration. During the field test, the RSM2 with the optimized drill bit design successfully achieved a DLS of 19.5 deg/100ft while utilizing 60% of the steering force on the ribs. Calculations indicate that it can deliver a DLS of 32 deg/100ft when using 100% of the steering force.

This concludes that the RSM2 in combination with an appropriate drill bit design, capable of providing a single solution for delivering the required directional response while managing tool availability and

flexibility across multiple BHA designs. Consequently, the number of trips could be reduced, which helps minimize flat time in drilling progress.

This novel synergy between drill bit design and RSM technology demonstrates a unique capability that helps achieve all drilling objectives with fewer trips, reduces flat time, manages the tool fleet, and delivers a wellbore with minimal tortuosity.

Introduction

The Rib-Steered Motor (RSM) represents a significant evolution in directional drilling technology, particularly within the context of Underbalanced Coiled Tubing Drilling (UBCTD). Unlike conventional 2 $\frac{7}{8}$ -in. Positive Displacement Motors (PDMs), which rely on a fixed bend and downhole orientation to steer the bit, the RSM is a closed-loop steerable drilling system adapted from rotary drilling technologies and optimized for coiled tubing operations. Its core innovation lies in the use of three hydraulically expandable ribs, spaced 120° apart, which are controlled from the surface via an electric line inside the coiled tubing. These ribs apply lateral force against the wellbore wall, enabling precise and real-time directional control without the need to slide or reorient the tool face.

This design offers several structural and operational advantages over traditional bent PDMs. The RSM integrates a mud motor for bit rotation, near-bit inclination and RPM sensors, and a compact 13-foot steering assembly, compared to the 18-foot length typical of conventional PDM setups. It operates in multiple modes: *Ribs Off* for tripping, *Inclination Hold* to maintain trajectory, *Steer Mode* for directional changes, and *Center Mode* to act as a stabilizer (Lallemant et al, 2025). These modes allow the RSM to drill both curves and tangents seamlessly, eliminating the need to trip out and reconfigure the bottom hole assembly (BHA) when transitioning between build and hold sections. This capability alone significantly reduces non-productive time (NPT) and simplifies BHA logistics.

One of the most impactful benefits of the RSM is its ability to maintain low Dogleg Severity (DLS) in lateral sections—typically between 2° and 5° per 100 feet—compared to the high DLS values (up to 35°/100 ft) often seen with bent PDMs. While high DLS is useful for aggressive build sections, it frequently results in tortuous well paths that increase friction, reduce weight transfer to the bit, and limit lateral reach. In contrast, the RSM's smoother, more linear trajectories significantly reduce drag and improve weight-on-bit efficiency, which is critical for maintaining high rates of penetration in extended laterals.

The surface-controlled steering of the RSM also enhances operational reliability by removing the dependency on downhole orientation tools, which are prone to failure and require frequent maintenance. This not only improves tool uptime but also reduces the complexity of the drilling system. Furthermore, the ability to steer while rotating improves hole cleaning and reduces the risk of differential sticking—common challenges in coiled tubing environments.

From a broader operational perspective, the RSM's design and functionality allow for a reduction in the number of BHAs required during a drilling campaign. Since the same tool can handle build, hold, and minor correction sections, operators can avoid multiple trips and tool changes, which translates to lower operational costs, faster well delivery, and reduced exposure to downhole risks. The improved wellbore quality also facilitates smoother casing or liner deployment and enhances long-term production performance by ensuring better reservoir contact.

In summary, the RSM's innovative design—featuring surface-controlled hydraulic ribs, integrated sensors, and multiple operating modes—delivers a step-change in directional drilling performance for UBCTD. It not only outperforms traditional bent PDMs in terms of trajectory control and operational efficiency but also enables longer, cleaner, and more productive lateral sections with fewer tools and less complexity.

Application Descriptions

This CTD application located onshore Middle East. It is targeting two different group of gas producer formations – named as Formation "A" and the formation below it, Formation "B" as illustrated in the Fig. 1.

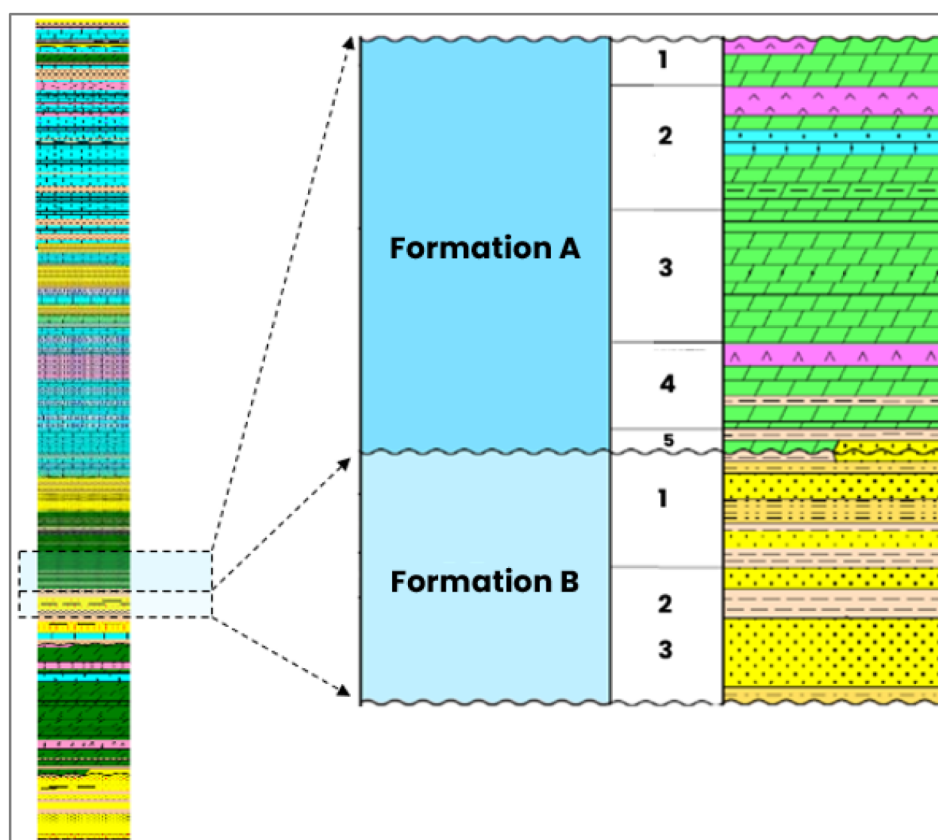


Figure 1—Main formations with common layers (Siddik et al, 2023)

The drilling campaign targets two distinct formations—Formation "A" in the upper section and Formation "B" in the lower section—each presenting unique lithological and mechanical challenges. The formations are part of a highly interbedded geological environment, where alternating layers of varying hardness and strength, from medium to hard, demand precise tool selection and operational strategies. The primary lithologies encountered are carbonates in Formation "A" and sandstones in Formation "B," each requiring different approaches to optimize drilling performance and mitigate risks.

Formation "A" is a carbonate-rich interval composed of five sub-layers, with drilling typically concentrated in sub-layers 2 and 3. These carbonates are hard and often unstable, leading to high vibration levels during drilling. Despite these challenges, Formation "A" has consistently delivered long drilling footage and relatively high rates of penetration (ROP), typically ranging from 15 to 20 ft/hr.

Historically, Positive Displacement Motors (PDMs) have been the preferred choice for drilling this formation. Their ability to deliver high torque at low rotational speeds has proven effective in managing the hard, brittle nature of the carbonates, while also minimizing bit wear. In many cases, bits were pulled out of hole (POOH) in re-runnable condition, highlighting the efficiency and durability of the PDM-based bottom hole assemblies (BHAs) in this environment.

In contrast, Formation "B" is composed of hard, abrasive sandstone and consists of three sub-layers. While more geomechanically stable due to its consolidated structure—resulting in lower vibration levels—the sandstone's abrasive nature leads to rapid wear of both the bit and BHA components. Drilling through this formation typically results in shorter footage and lower ROP, and bits often return to surface in severely

worn or irreparable condition. The high friction between the steel sleeve and the borehole wall also generates elevated temperatures, further accelerating tool wear and reducing operational efficiency. Turbine BHA used to drill this formation successfully in this campaign.

The historical success of PDMs in Formation "A" provides a strong foundation for the introduction of Rib-Steered Motor (RSM) technology in this application. The PDM's proven ability to handle the formation's instability and vibration-prone conditions makes it a natural partner for RSM systems, which offer enhanced directional control and continuous rotation.

By integrating RSM with PDMs, operators can maintain the mechanical advantages of the motor while gaining the steering precision and real-time trajectory adjustments offered by RSM. This hybrid approach is particularly valuable in carbonate formations where borehole instability, high differential pressures, and the risk of stuck BHAs are prevalent.

In summary, the operational history in Formation "A" demonstrates that PDMs are well-suited for managing the challenges of carbonate drilling. Building on this foundation, the application of RSM technology—leveraging the torque and stability of PDMs—represents a logical and effective evolution. This combination not only enhances drilling efficiency and bit life but also improves wellbore quality and trajectory control, making it a compelling solution for future campaigns targeting similar carbonate environments.

Coiled Tubing Drilling Sequence for Multi-Lateral Wells

This application utilizes coiled tubing BHAs to drill multiple laterals in a single well, following the sequence illustrated in Fig. 2.

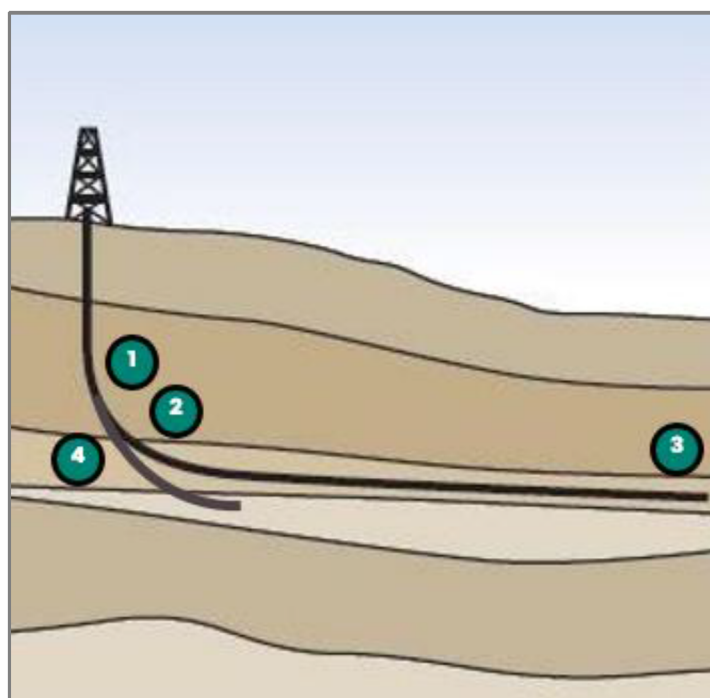


Figure 2—Operation sequence on coil-tubing drilling (Siddik et al, 2023)

Clean-Out Operation

Drilling begins with a clean-out run using a medium-AKO PDM BHA to drill out cement, the shoe (or reamer shoe), and the rathole.

Build-Up Section

Once the wellbore is clear, a high-AKO PDM BHA is used to drill the directional build-up section. This section is typically short and ends once the planned inclination and azimuth are achieved.

Lateral Section

After completing the build-up, the BHA is pulled out and replaced with a medium- or low-AKO BHA to drill the lateral section. Each well is typically planned for three laterals, though additional laterals may be added during operations. Drilling through the "A" formation often requires one bit per lateral, though bits may be reused depending on wear.

Side-Tracking for Additional Laterals

Upon reaching TD of a lateral, the same BHA and bit may be used to initiate an open-hole sidetrack (OHST) for the next lateral. If OHST fails due to bit dullness or AKO limitations, a trip is required to run a high-AKO BHA for re-orientation, followed by another trip to switch back to a medium-AKO BHA to continue drilling.

First Generation RSM Deployment in the Application

The deployment of the Rib-Steered Motor (RSM) in Well-C marked a significant leap in drilling efficiency and trajectory control. With a total of 7,875 ft drilled and 6,921 ft of reservoir contact, the RSM enabled the drilling of the longest single lateral section at 3,172 ft and achieved a record 1,350 ft drilled in a single calendar day. Most notably, the well—originally scheduled for 42 days—was completed in just 24 days, saving 18 days or 42.8% of operational time (Kashif et al, 2012).

This efficiency gain was largely due to the RSM's ability to steer while rotating, maintain low dogleg severity (DLS), and deliver a smoother, more linear wellbore trajectory. The top view of Lateral-1, drilled with a PDM, shows a tortuous path, while Lateral-2 and 3, drilled with the RSM, exhibit clean, straight profiles—highlighting the RSM's superior directional control and its ability to reduce friction and improve weight transfer to the bit as shown in Fig. 3.

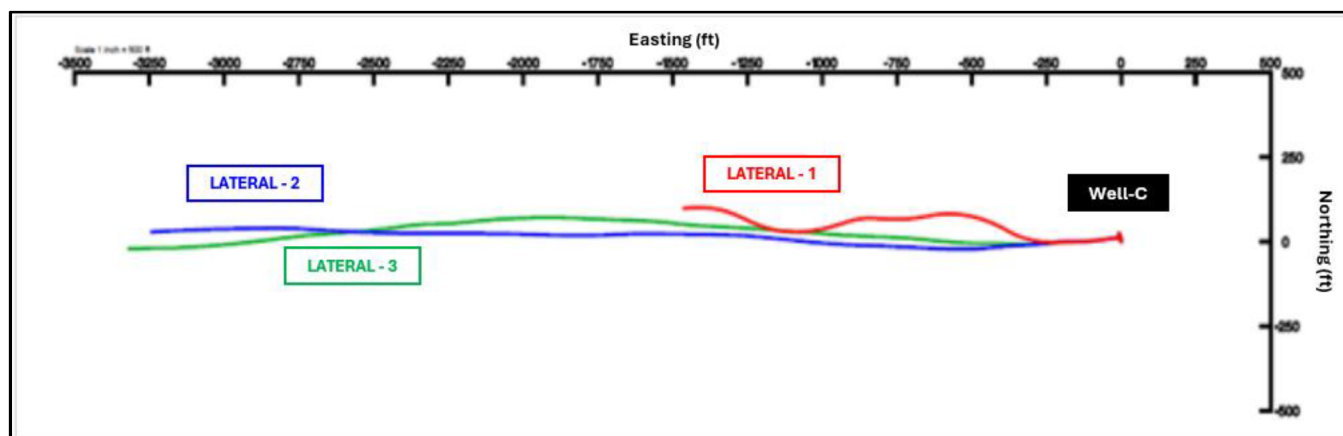


Figure 3—RSM Drilled Well (Ahmed et al, 2012)

In contrast, Well-D, drilled using a conventional 2 $\frac{7}{8}$ -in. PDM, achieved a slightly higher total drilled footage of 8,025 ft and 6,827 ft of reservoir contact. However, the well took 38.5 days to complete against a 43-day plan, yielding only a 10.5% of time savings. The well path image for Well-D reveals a tortuous trajectory, typical of PDM-drilled laterals, which can increase drag, reduce drilling efficiency, and complicate casing or liner deployment as shown in Fig. 4.

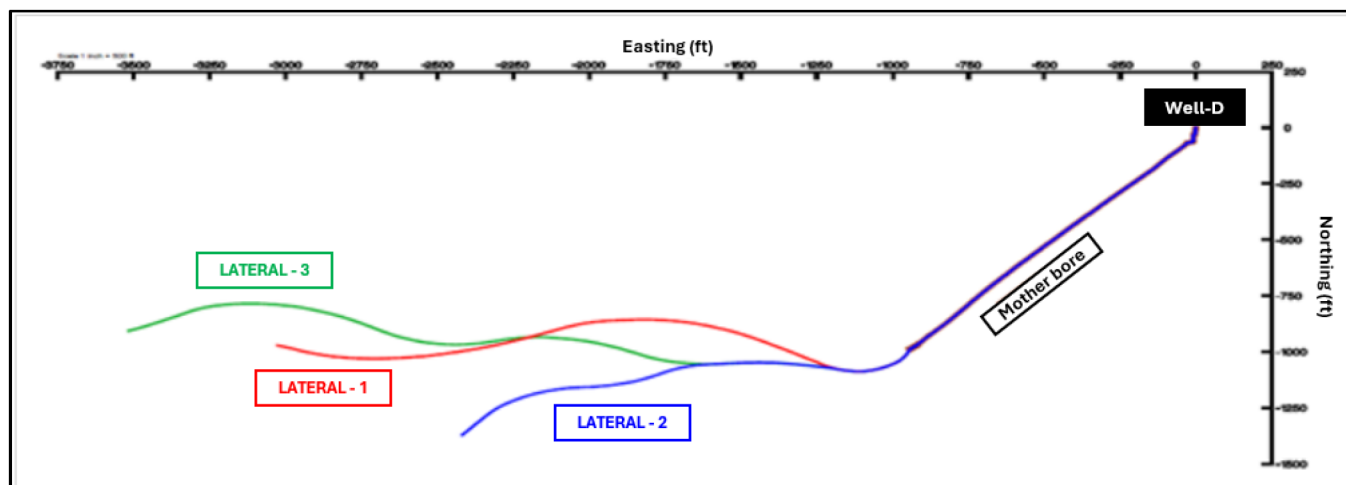


Figure 4—RSM Drilled Well (Ahmed et al, 2012)

The PDM's reliance on sliding for directional control and its point-the-bit steering mechanism often results in inconsistent wellbore geometry, especially in extended-reach or re-entry wells. This comparison clearly illustrates the RSM's advantage in delivering smoother wellbores and faster drilling performance in suitable applications.

The RSM's closed-loop, push-the-bit steering system – featuring three hydraulically expandable ribs controlled from the surface – enables real-time directional adjustments without interrupting rotation. This not only improves hole cleaning and reduces the risk of differential sticking but also enhances drilling efficiency in complex well architectures. Its compact design, integrated sensors, and multiple operating modes (such as inclination hold and steer mode) further contribute to precise trajectory control and reduced need for multiple BHAs as illustrated in Fig. 5.

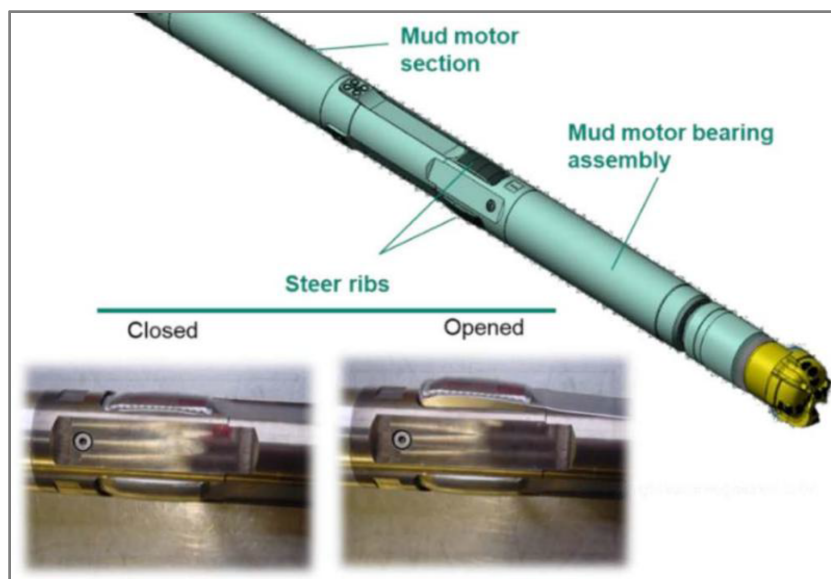


Figure 5—Rib-Steered Motor (RSM) Design (Funes et al, 2025)

The first-generation Rib-Steered Motor (RSM) introduced a novel approach to directional drilling, but it came with notable limitations. Its build-up rate was capped at 10°/100 ft, which constrained its use in high-DLS (dogleg severity) applications. Additionally, its temperature tolerance was limited to 150°C, making it less suitable for high-temperature reservoirs.

Another critical constraint was its dependency on bit design—specifically, the RSM required a bit tailored for push-the-bit systems. Using a bit optimized for positive displacement motors (PDMs) often led to suboptimal steering response and reduced overall performance, highlighting the need for a more integrated design approach as shown in Fig. 6.

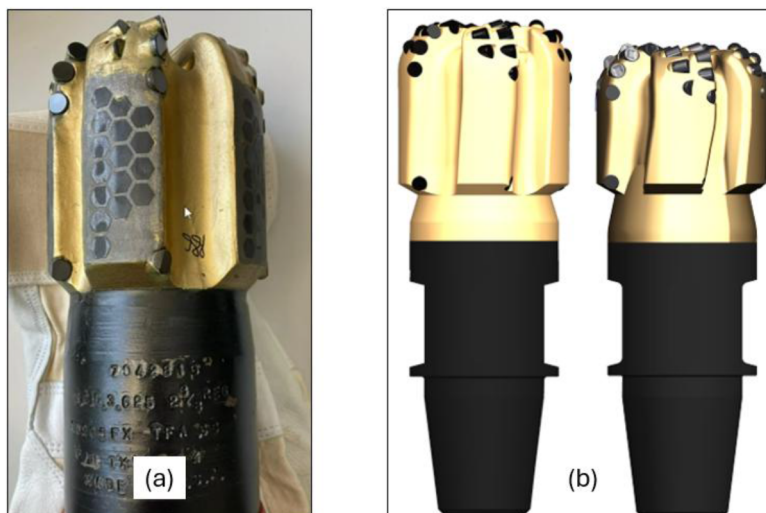


Figure 6—5 blades and 8 mm fixed cutters PDC bit: (a) design used for test; (b) design feature variety.

To overcome these challenges, the second-generation RSM (RSM2) was developed with significant enhancements in both mechanical and thermal performance. RSM2 features an improved build rate capability of up to 25°/100 ft and upgraded electronics and hydraulics rated for 160°C, enabling it to operate reliably in more demanding environments.

A major milestone in its development was a lab test conducted using a sandstone block, where RSM2 successfully executed an open-hole sidetrack with a build rate of 27°/100 ft—exceeding its design target (Fig. 7). This lab validation not only confirmed the tool's steering precision and durability but also demonstrated the effectiveness of the DLS platform in replicating complex downhole conditions for performance evaluation.

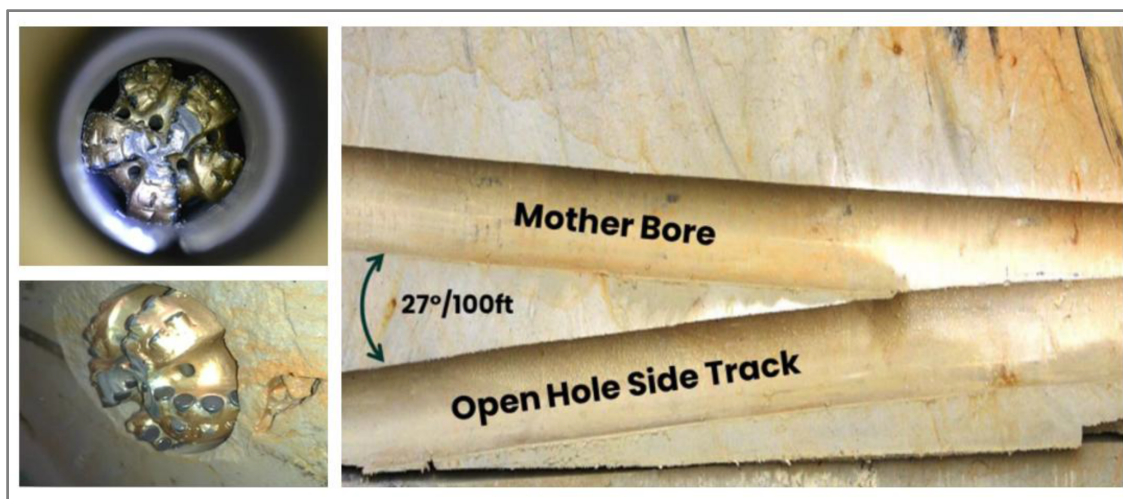


Figure 7—Cut view of a sandstone block showing an open hole sidetrack with RSM2 (Lallemant et al, 2025)

Despite the significant advancements in RSM2's mechanical and thermal capabilities, its success in the field remains closely tied to drill bit compatibility. The RSM operates on a push-the-bit steering principle, similar to conventional rotary steerable systems (RSS), which demands a bit specifically engineered to complement this mechanism. Using a bit optimized for point-the-bit systems, such as those designed for positive displacement motors (PDMs), can severely compromise steering response, trajectory control, and overall drilling efficiency.

To fully leverage RSM2's high build rate and directional precision, a dedicated bit design strategy is essential—one that aligns with the tool's steering dynamics and operational goals across various formations and well profiles.

Unlike PDMs, which steer by orienting the bit itself, push-the-bit systems like RSM2 rely on generating side forces through the tool body. This requires bits with tailored cutter layouts that promote lateral force transmission, gauge pad geometries that support stable wall contact, and vibration mitigation features to preserve directional control. When properly matched with such optimized bit designs, RSM2 demonstrates measurable gains in drilling speed, trajectory smoothness, and operational efficiency—especially in wells requiring extended laterals and high dolog severity. The synergy between tool and bit is not just a performance enhancer; it is a prerequisite for unlocking the full potential of RSM2 in complex drilling environments.

Drill Bit Design Methodology

Due to the specific nature of the application, optimal bit selection was a critical factor for successful execution. In the absence of analytical precedent to establish reliable assumptions regarding bit behavior in conjunction with the RSM2, a tailored methodology was developed (Fig. 8). This approach integrated state-of-the-art simulation software with expert evaluations from bit and directional drilling specialists, aiming to define quantitative parameters to guide bit design characteristics suitable for the drilling environment.



Figure 8—Methodology for Bit Design Research and Development

Bit Response Evaluation

Once the specific application conditions, drilling environment, and bottom hole assembly (BHA) response were defined, the verification of bit behaviors influencing typical bit design was conducted. These behaviors were classified into five categories: stability, aggressiveness, durability, steerability, and hydraulics. Based on prior experience with downhole motors in offset wells, the new bit design focused primarily on steerability, stability, and aggressiveness.

Analysis

The analysis was structured around outputs generated by proprietary software: the Directional Behavior Simulator and the 3D Drill Bit Behavior Simulator. These tools were used to evaluate lateral aggressiveness (build-up rate capability) and axial aggressiveness (weight on bit and torque fluctuations).

Directional Behavior Simulator

To establish a baseline for the 3.625" section analysis, a PDC bit with 5 blades and 8 mm fixed cutters was selected (Fig. 9). This bit had demonstrated excellent performance in offset wells. The simulation setup was based on real field data, including lithological descriptions, mud weight, drilling parameters (RPM, ROP, WOB, torque, directional tool forces), bit design, and BHA geometry.

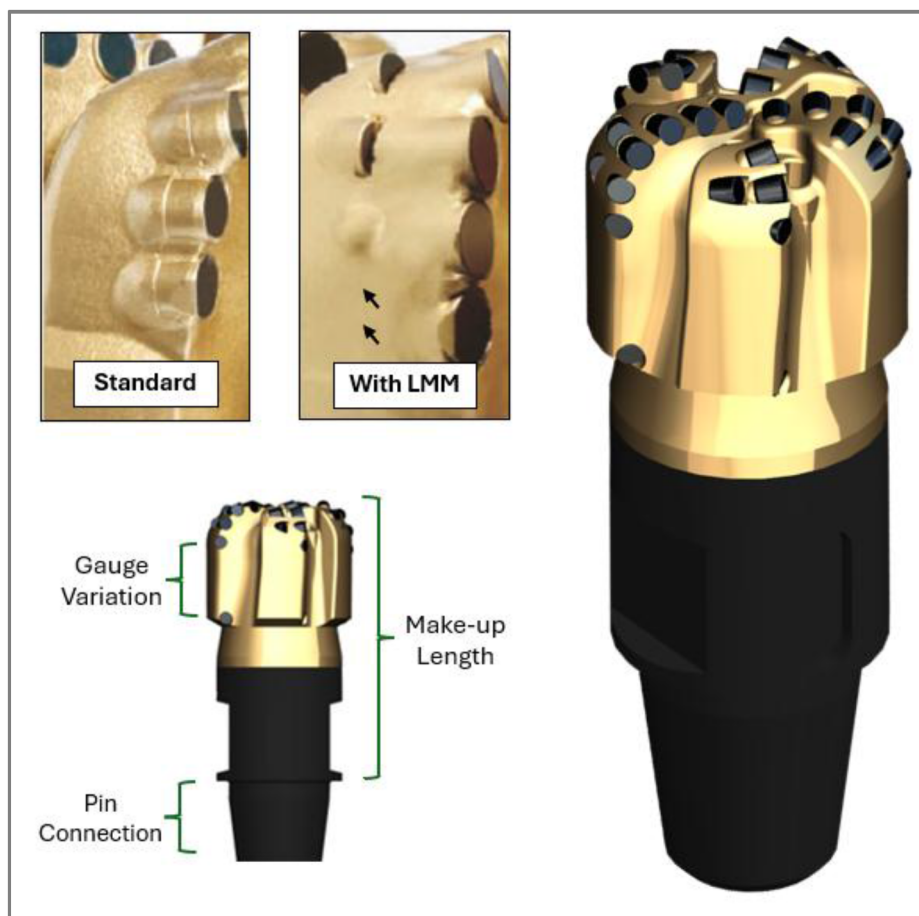


Figure 9—Generic drill bit showing features affecting directional response and vibration mitigation.

Over 30 iterations were conducted to understand, calibrate, and verify bit behavior. The following cases summarize the key findings.

Case #1: Gauge Variation and Steer Force Impact

The first objective was to analyze Build-Up Rate (BUR), Dog Leg Severity (DLS), and Bit Side Force response while maintaining a constant gauge length and varying the under-gauge by 0.03", 0.05", 0.07", and 0.09". These variations were tested across different Steer Force (SF) percentages (50%, 75%, and 100%) applied by the directional tool.

The graphical results helped prioritize considerations regarding BUR in relation to steer force demands, contributing to a more informed selection process that accounted for potential impacts on DLS as shown on Fig. 10.

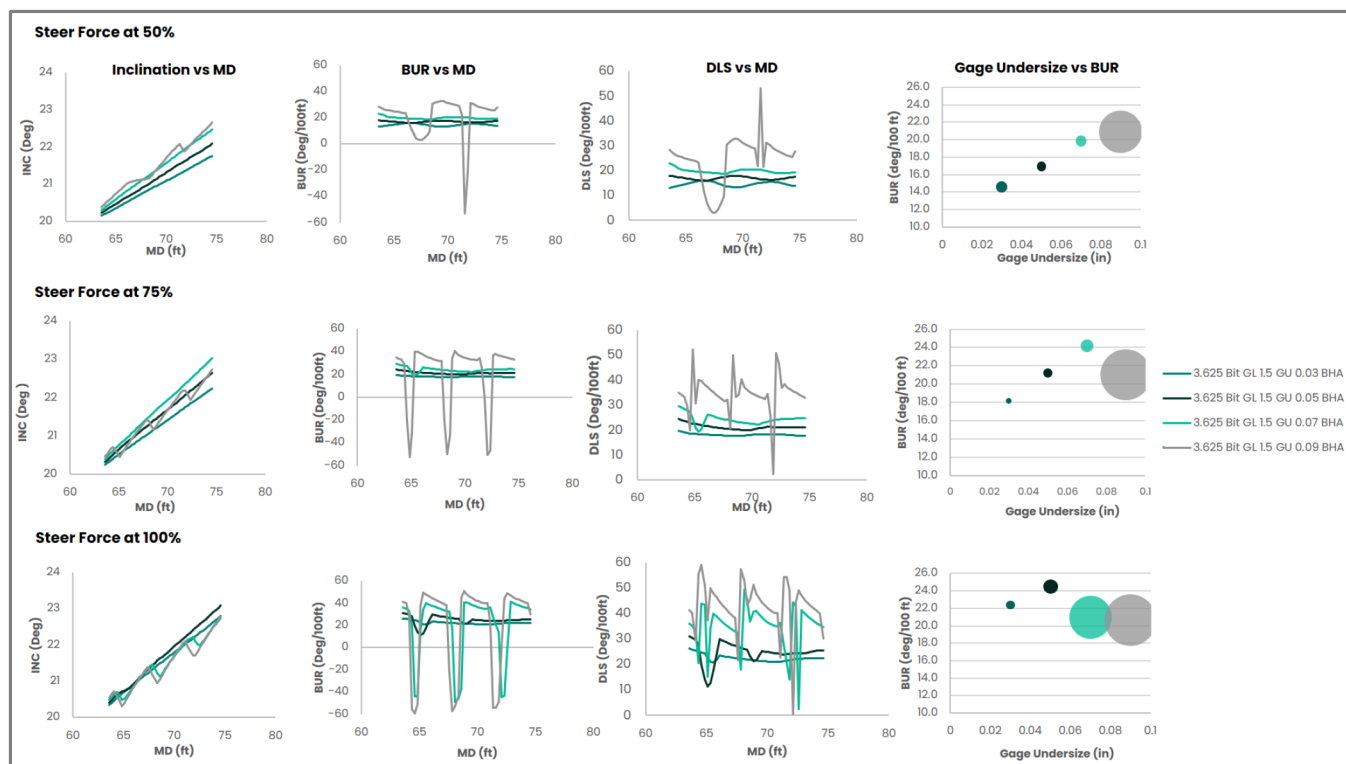


Figure 10—Iteration – baseline drill bit design.

Case #2: Blade and Cutter Size Comparison

This scenario aimed to provide a quantitative comparison using a PDC bit with 6 blades and 11 mm cutters vs incumbent design with 5 blades and 8 mm cutters, under identical input parameters. The insights gained supported the selection of the appropriate PDC bit frame for future iterations as shown on Fig. 11.

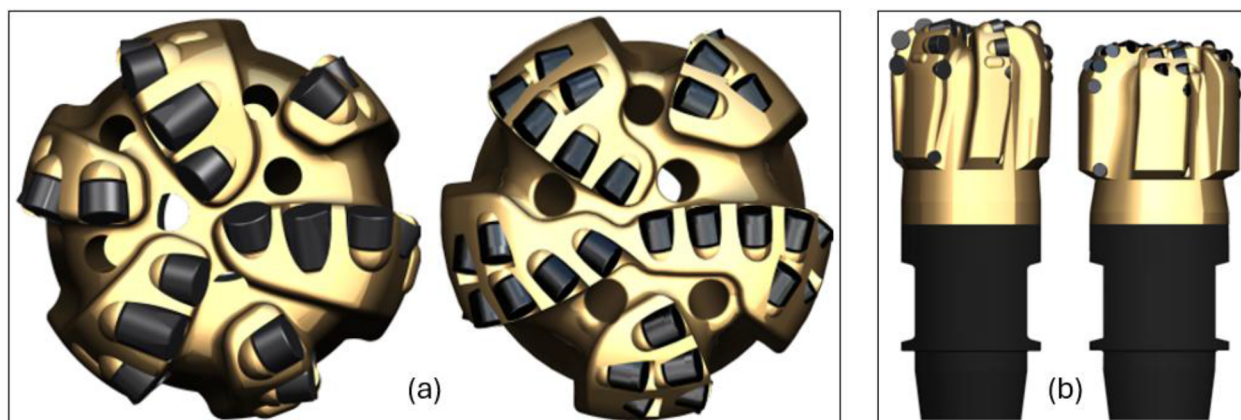


Figure 11—6 blades and 11 mm cutters (left) and 5 blades and 8 mm cutters (right) PDC bit: (a) face view; (b) side view.

Case #3: Make-Up Length (MUL) Influence

To assess the influence of bit geometry on directional response, simulations were conducted by varying the Make-Up Length (MUL) while keeping the gauge length constant. A 75% Steer Force was applied across different under-gauge values. The results guided the selection of the optimal MUL for the application as shown on Fig. 12.

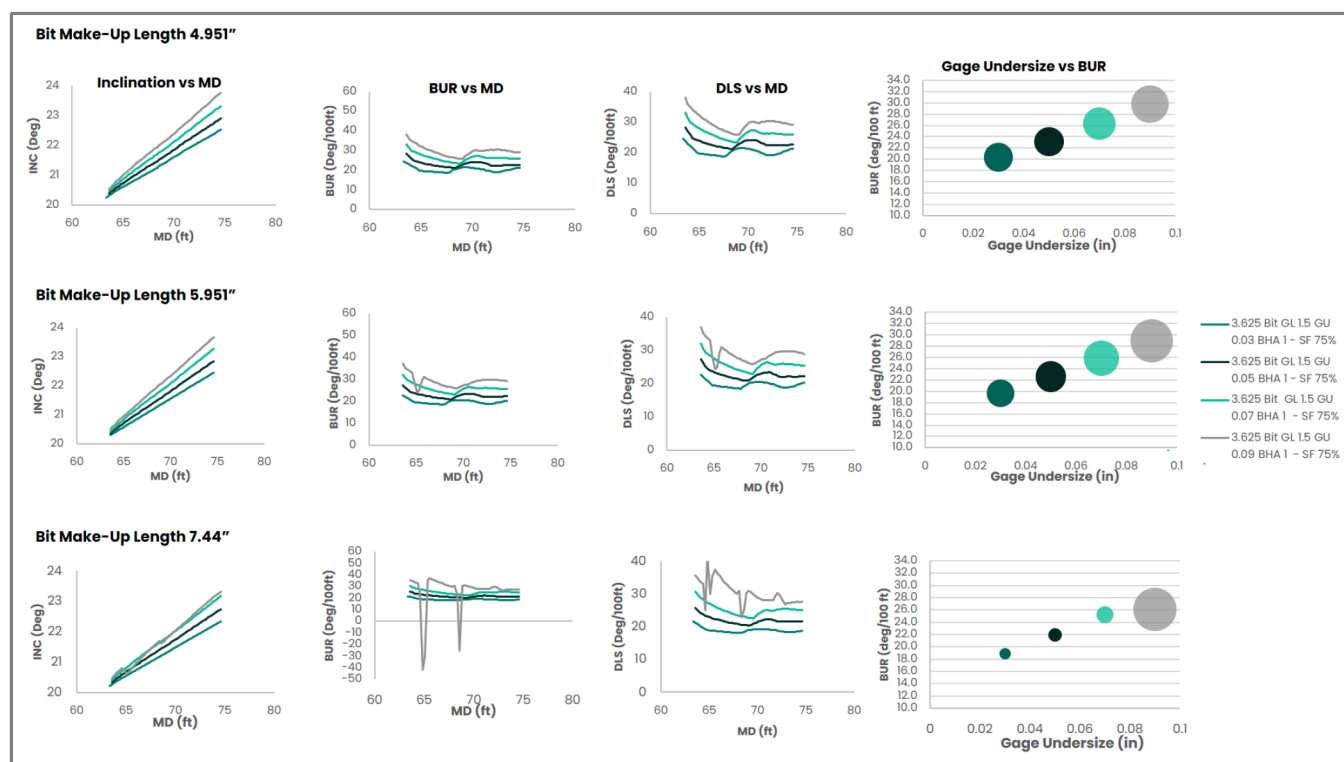


Figure 12—Iteration – make-up length (MUL)

3D Drill Bit Behavior Simulator

In the second phase of the analysis, the focus shifted to evaluating the performance of a 5-blade, 8 mm fixed cutter bit under varying Lateral Movement Mitigator (LMM) configurations using advanced drilling simulation software. These high-fidelity simulators are instrumental in modeling the complex interactions between bit design, formation properties, and BHA dynamics. Leveraging the Directional Behavior Simulator, a new bit design was developed incorporating previously quantified improvements. Two sub-variants were created—featuring LMM values of 0.015" and 0.030"—to assess their directional performance using the 3D Drill Bit Behavior Simulator.

Drilling simulation software plays a pivotal role in advancing automation, control, and optimization of drilling operations. These tools model complex interactions between surface and downhole systems, drilling fluids, and geological formations. While full-system modeling remains challenging due to uncertainties and dynamic conditions, high-fidelity simulations in areas like rig mechanics, bit-rock interaction, and downhole dynamics have shown promising results. Adaptive simulations are increasingly used to predict and manage downhole behavior in real time, supporting intelligent managed pressure drilling and enhancing operational decision-making. The technology continues to evolve toward fully integrated, predictive models that can simulate entire drilling processes with greater accuracy and reliability (Russell et al, 2022)

This simulator enables precise control over a wide range of input parameters, including rock type, unconfined compressive strength (UCS), BHA misalignment, and subtle bit geometry changes (Fig. 13). Such flexibility allows engineers to simulate realistic downhole conditions and predict bit response with high accuracy. The iterative simulation process provided critical insights into how minor design modifications influence directional tendencies, ROP, and overall drilling efficiency. As a result, the optimized bit design was finalized to deliver improved directional control tailored to the specific challenges of the application.

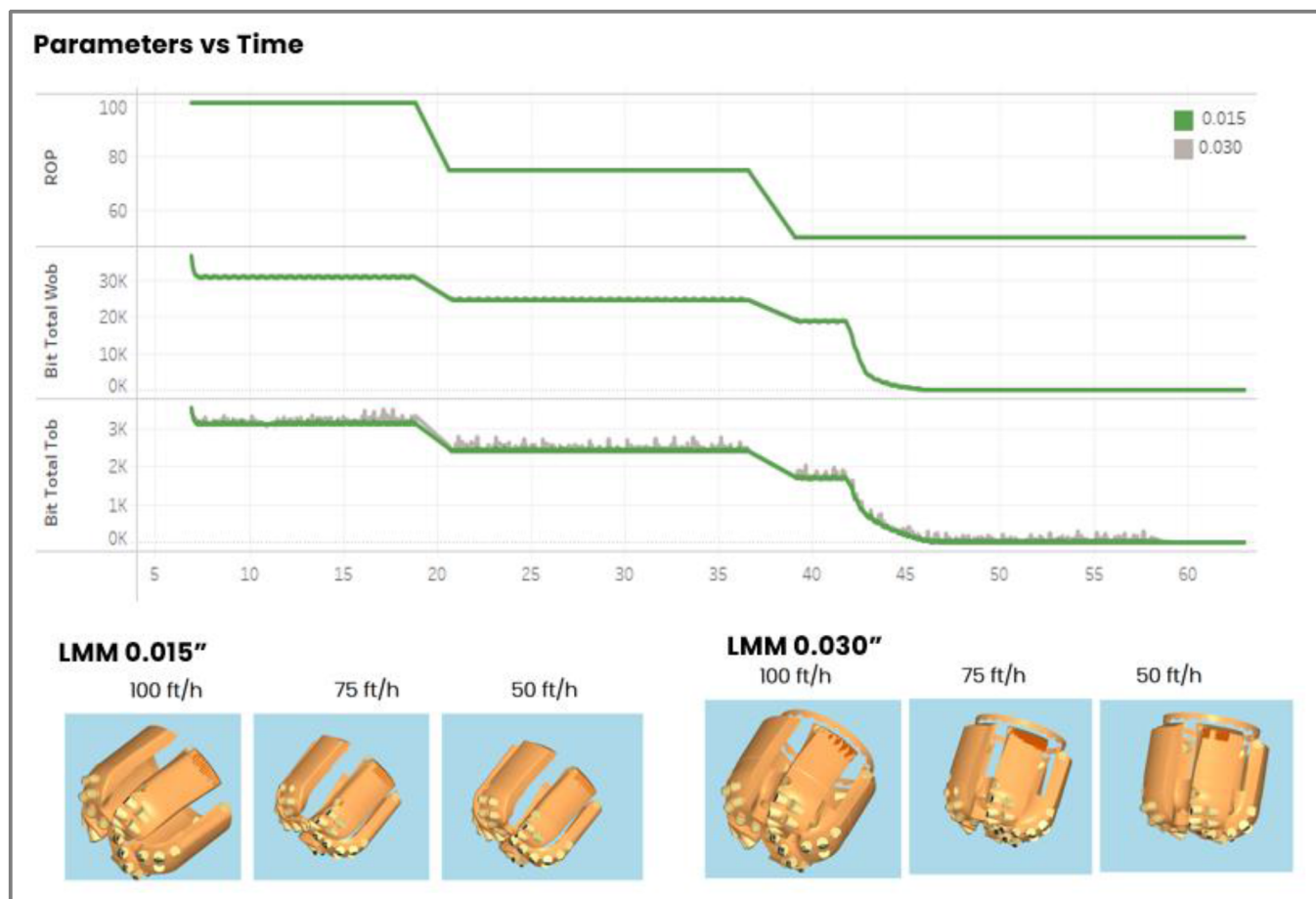


Figure 13—3D Drill Bit Behavior Simulator – LMM Results.

Field Result

The Rib-Steered Motor 2 (RSM2) underwent a comprehensive field evaluation across two drilling runs, demonstrating its advanced directional control and high dogleg capability. In Run 1 (Fig. 14), the tool initiated at an inclination of 74.23° and successfully built up to 89°, while executing a turn from 50° to 60° azimuth. The build section maintained an average dogleg severity (DLS) of 12.41°/30m, closely aligning with the planned trajectory. Notably, during the landing phase at 85° inclination, the RSM2 achieved a right turn with a DLS of 19.53°/100ft, showcasing its ability to deliver aggressive curvature while maintaining trajectory control and borehole stability.

Throughout the build-up section in Run 1, the RSM2 consistently outperformed legacy RSM, delivering peak DLS values exceeding 16°/100ft. In the lateral section, the tool achieved DLS values of 19.49° and 19.88°/30m while executing turns, using only 60% and 50% of its available steer force, respectively. These results indicate that the RSM2 has the potential to reach or exceed a DLS of 25°/100ft, a significant improvement over the previous generation tools, which maxed out at 11°/100ft. This performance leap reflects advancements in motor design, steering efficiency, and tool responsiveness.

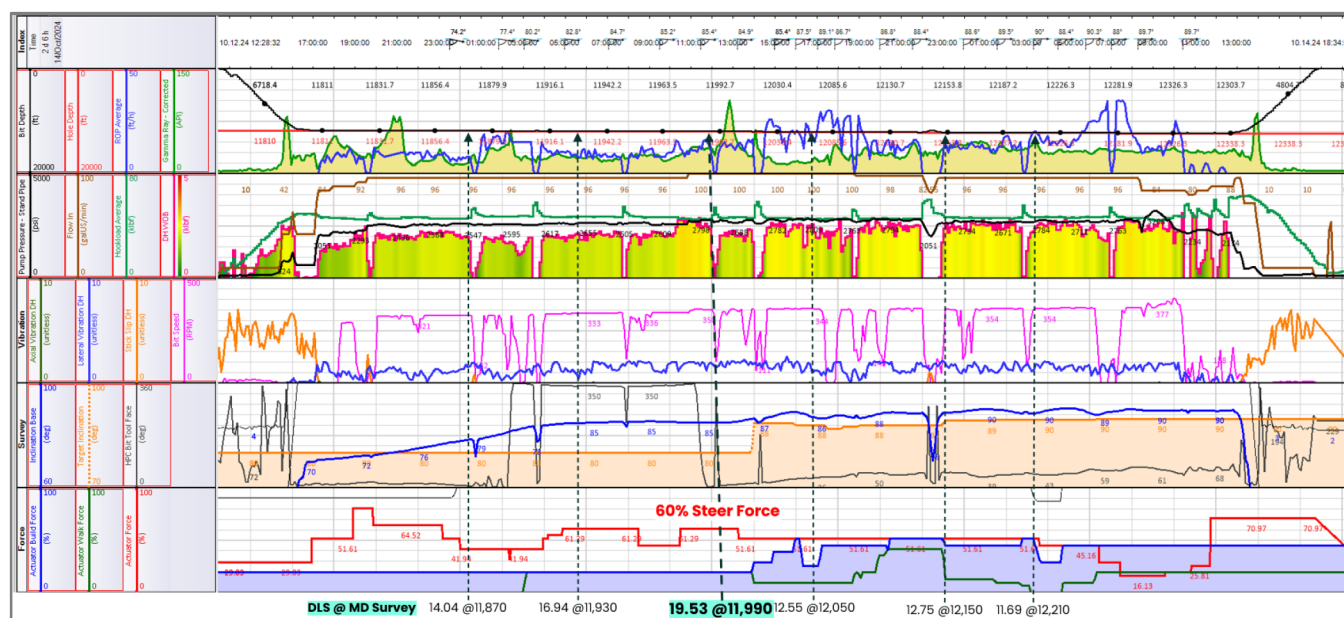


Figure 14—RSM2 Run 1 Overview

Run 2 (Fig. 15) further validated the RSM2's capabilities, particularly in complex directional applications. At 12,447 ft MD, the tool executed a turn from 70° to 60° azimuth with a DLS of 19.88°/30m, again using only 50% steer force. This level of performance with minimal input highlights the tool's efficiency and suitability for high-angle and horizontal drilling, especially in formations requiring aggressive build rates and precise azimuthal control. The tool's ability to maintain consistent trajectory while minimizing tortuosity is a key advantage in reducing drag, improving hole quality, and enhancing overall drilling efficiency.

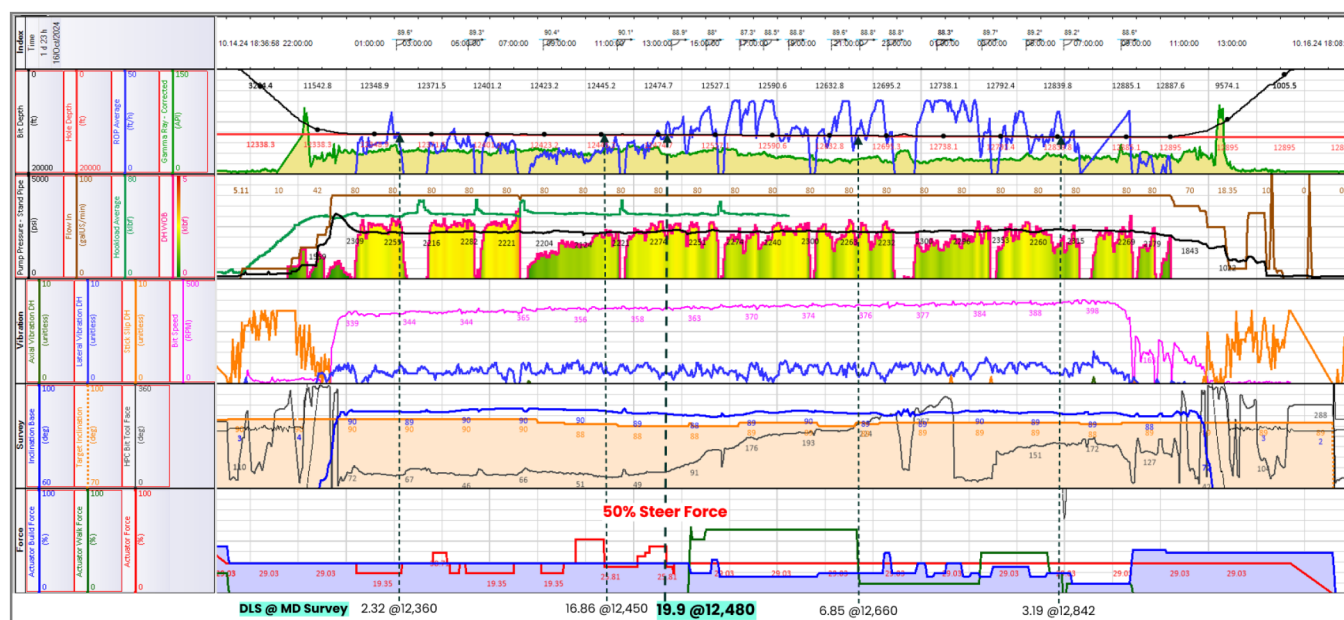


Figure 15—RSM2 Run 2 Overview

When compared to a conventional PDM BHA used in Run 3, the RSM2 delivered a noticeably smoother well path with improved directional control. The PDM, while historically effective in carbonate formations like Formation "A," lacks the real-time steering precision and continuous rotation capability of the RSM2.

However, the historical success of PDMs in Formation "A" is critical to understanding the RSM2's relevance.

PDMs have long been favored for their torque delivery and ability to handle hard, unstable formations, and their integration with RSM technology leverages these strengths while adding superior steering control. This hybrid approach is particularly effective in interbedded formations where both directional agility and mechanical robustness are required.

A crucial aspect of validating the RSM2's performance lies in the analysis of bit wear. Post-run inspections revealed wear patterns consistent with push-the-bit steering behavior, particularly in areas subjected to side-cutting during directional maneuvers. The wear was concentrated on the gauge and shoulder areas of the bit, where increased contact with the formation occurred due to the applied steer force.

This wear pattern aligns with the expected mechanical interaction in carbonate formations, confirming that the bit was effectively engaged in directional work without excessive or uneven wear that could compromise performance or reusability.

The comparison between bits used with RSM2 and those used with conventional PDMs further reinforces the tool's effectiveness. While PDM-driven bits typically show more uniform wear due to point-the-bit behavior, the RSM2 bits exhibited targeted wear in high-contact zones, validating the tool's steering action and confirming that the bit design was appropriate for the application as shown in Fig. 16.

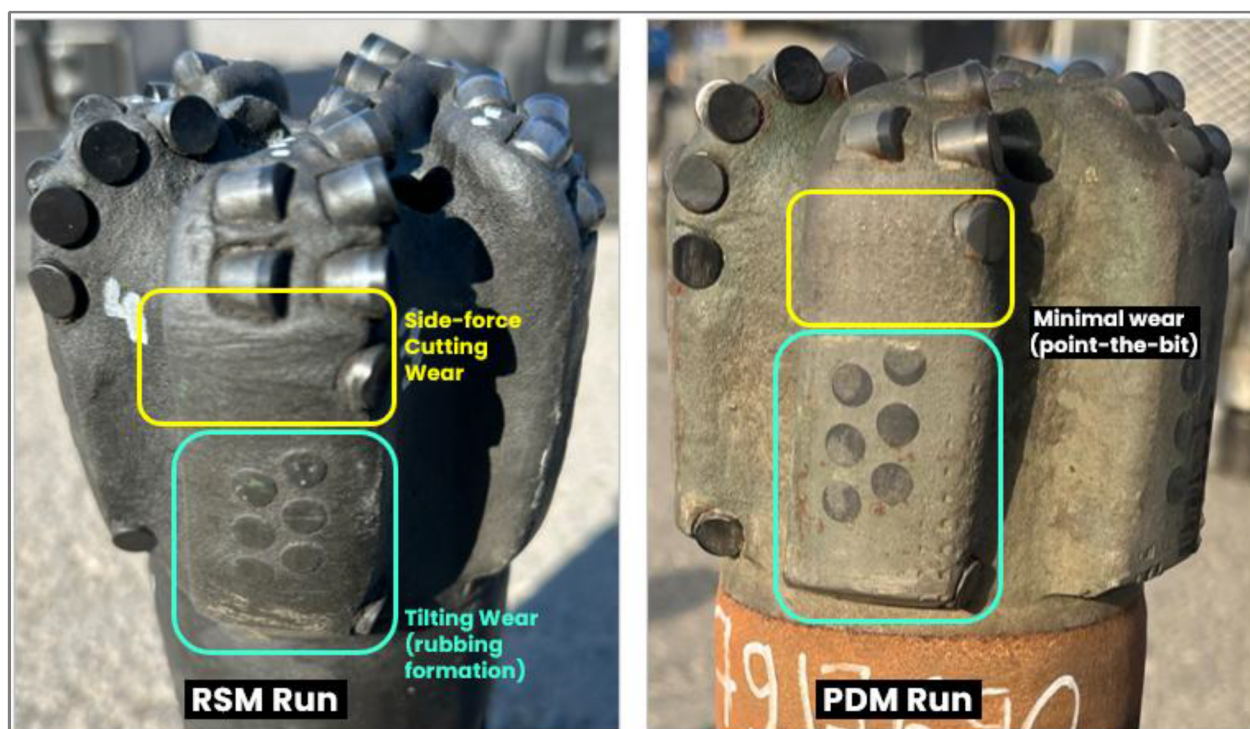


Figure 16—Bit wear trend between RSM and PDM after interaction with formation.

The ability to correlate bit dull condition with performance metrics such as DLS and ROP provides valuable feedback for optimizing bit selection and BHA design in future runs. This alignment between bit wear and directional performance not only confirms the RSM2's operational success but also supports its continued deployment in challenging carbonate environments.

Conclusion

The evolution from the first-generation Rib-Steered Motor (RSM) to second-generation, RSM2 marks a significant leap in directional drilling technology. RSM2 addressed key limitations of its predecessor

by increasing the build-up rate capability to 25°/100 ft and improving thermal tolerance to 160°C. Field tests confirmed its superior steering control, achieving high DLS with reduced steer force, and delivering smoother well paths compared to conventional PDM BHAs.

A critical factor in this success was the integration of a simulation-driven bit design methodology, which optimized lateral movement mitigators and cutter placement to align with the push-the-bit steering mechanism, resulting in improved ROP and predictable wear patterns.

Looking ahead, RSM3 is under development to push operational boundaries further, with a target temperature rating of 175°C and enhanced build rate performance. It will incorporate Multi-Chip Module (MCM) electronics and align with third-generation CTD BHAs for improved reliability in extreme environments (Lallemand et al, 2025). Continued use of advanced bit behavior simulators will ensure that bit designs are precisely tailored to formation and tool dynamics. Field validation, particularly through bit dull analysis, will remain essential to confirm performance alignment and steerability, ensuring the next generation of RSM technology delivers even greater efficiency, durability, and adaptability.

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References

- Ahmed, K., Kirby, C., Law, A., and Al-Humood, M. 2012. Improved Technology Enhances Gas Production and Decreases Cost In Underbalanced Drilling While Using Coiled Tubing Directional Drilling. Presented at the SPE Saudi Arabia Section Technical Symposium and Exhibition, Al-Khobar, Saudi Arabia, April. SPE-160871-MS. <https://doi.org/10.2118/160871-MS>
- Funes, D., Lallemand, M., and Shaikh, M. 2025. Slim Hole Drilling: Bridging the Gap Between Oil and Gas Demand and a Sustainable Energy Future. Presented at the Offshore Technology Conference, Houston, Texas, USA, 28 April. OTC-35987-MS. <https://doi.org/10.4043/35987-MS>
- Kozlov, A., Frantzen, S., Gorges, T., Al Khamees, S., Guzman, J., and Aduba, A. 2010. Next Generation Technologies for Underbalanced Coil Tubing Drilling. Presented at the SPE Deep Gas Conference and Exhibition, Manama, Bahrain, 24–26 January. SPE-132084-MS. <https://doi.org/10.2118/132084-MS>
- Lallemand, M., Funes, D., and Shaikh, M. 2025. Thirty Years of Successful Coiled Tubing Re-Entry Drilling to Enhance Production in Mature Assets. Presented at the GOTECH, Dubai City, UAE, April. SPE-224810-MS. <https://doi.org/10.2118/224810-MS>
- Ross, M., Anyanwu, O.N., Klotz, C., and Ulrich, C.D. 2012. Rib-Steered Motor Technology: The Revolutionary Approach Extends the Coiled Tubing Drilling Application Scope. Presented at the SPE/ICoTA Coiled Tubing & Well Intervention Conference and Exhibition, The Woodlands, Texas, USA, 27 March. SPE-153573-MS. <https://doi.org/10.2118/153573-MS>
- Russell, S. Craig, Duffy, Stephen, and Matthews, Oliver. 2022. Shaped Cutter Performance Optimization Through Advanced Drilling Simulations. Paper presented at the IADC/SPE International Drilling Conference and Exhibition, Galveston, Texas, USA, March. SPE-208681-MS. <https://doi.org/10.2118/208681-MS>
- Siddik, S. M., Rojas, P., Quintero Barrera, J. L., and Alraheb, M. 2023. Coiled Tubing Drilling Advancement with Smallest Shaped Diamond Cutting Element. Presented at the Middle East Oil, Gas and Geosciences Show, Manama, Bahrain, 19–21 February. SPE-213666-MS. <https://doi.org/10.2118/213666-MS>